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What is the use of /etc/hosts file in Linux?

Pick **ONE** option



It contains the list of all users.



It contains a list of all files in the system



It contains the time and date information.



It contains the FQDN to IP address mappings.

Clear Selection

Which Clause restricts the groups of rows returned to those groups for the



11

Pick **ONE** option

☐ Where clause

12

☐ Having Clause

13

☐ Distinct

14

☐ Exists

15

Clear Selection

16

Which interrupt is defined for system calls in Linux?

Pick **ONE** option

☐ 0x10

☐ 0x60

☐ 0x80

☐ 0x40

21. Which of the following is false?

Which of the following is false?

Pick **ONE** option

☐ 2-3 tree requires less storage than the BST

☐ lookup in 2-3 tree is more efficient than in BST

☐ 2-3 tree is shallower than BST

☐ 2-3 tree is a balanced tree

20. In what way the Symmetry Sparse Matrix can be stored efficiently?

In what way the Symmetry Sparse Matrix can be stored efficiently?

Pick **ONE** option



Heap



Binary tree



Hash table



Adjacency List

Clear Selection

ALL

15

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19. What is the purpose of ARP (Address Resolution Protocol) in computer networking?

What is the purpose of ARP (Address Resolution Protocol) in computer networking?

Pick **ONE** option

☐ To convert IP addresses to MAC addresses

☐ To convert domain names to IP addresses

☐ To establish a secure connection between client and server

☐ To route packets between networks

Clear Selection

14 15 16 17 18 19 20

18. In a ring topology network, if one device fails, how does it affect the network?

In a ring topology network, if one device fails, how does it affect the network?

Pick **ONE** option

☐ Only the failed device is affected

☐ All devices in the network are affected

☐ Only the devices between which the failed device exists are affected

☐ The network continues to function normally

Clear Selection

17. Which is the HTTP request method used to retrieve the metadata associated

Which is the HTTP request method used to retrieve the metadata associated with the resource's state?

Pick **ONE** option

☐ GET

☐ PUT

☐ HEAD

☐ POST



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What is the use of /etc/hosts file in Linux?

Pick **ONE** option



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It contains the FQDN to IP address mappings.

Clear Selection

What does idempotency mean in REST API design?

Pick **ONE** option

- ☐ The ability of an API to handle multiple requests in parallel
- ☐ The property that a method can be called multiple times without
- ☐ The capability of an API to update data
- ☐ The feature of an API that allows it to delete resources

Clear Selection

3. Operating system maintains the information about a process in

Operating system maintains the information about a process in which of the following?

Pick **ONE** option

☐ Heap

☐ Process Control Block

☐ Stack

☐ TLB Cache

Which Clause restricts the groups of rows returned to those groups for the



11

Pick **ONE** option

☐ Where clause

12

☐ Having Clause

13

☐ Distinct

14

☐ Exists

15

Clear Selection

16

What is the postfix expression for the corresponding infix expression? $a+b*c+(d*e)$

Pick **ONE** option

☐ $abc*+de*+$

☐ $abc+*de*+$

☐ $a+bc*de+*$

☐ $abc*+(de)*+$

Clear Selection

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

- ☐ This memory address does not map to actual memory cell that corresponds to address number.
- ☐ This memory is the same as physical memory.
- ☐ This memory cannot be accessed by applications at run time.
- ☐ This memory address is always present even after a system is restarted.

near Selection

Which of the following design criteria are not valid/suitable for Non-Relational Databases?

Click **ONE** option

☐ Your application required super-low latency

☐ Your data are unstructured without any relation between them

☐ You only need to serialize & deserialize data (JSON, XML, YAML)

☐ You need to store a very small amount of data

7. Consider the below java method. What's the output of this method? public static void function_1(int [] data, int low, int high) { int temp = data[low]; data[low] = data[high]; data[high] = temp; function_1(data, low + 1, high - 1); }

Consider the below java method. What's the output of this method?

```
public static void function_1(int [] data, int low, int high) {  
    if (low < high) {  
        int temp = data[low];  
        data[low] = data[high];  
        data[high] = temp;  
        function_1(data, low + 1, high - 1);  
    }  
}
```

ONE option

) The sum of all the elements of the array

What does CAP theorem describe with respect to System Design ?

Pick **ONE** option

☐ The relationship between CPU and memory usage

☐ The trade-offs between consistency, availability, and partition tolerance

☐ The process of data encryption

☐ The principles of object-oriented programming

25. Favorite Color

There are five people: Jack, Matthew, Albert, Peter, and Sebastian.
Each plays one sport out of football, cricket, volleyball, badminton, and squash.
They are from South Korea, the UK, India, China, and Russia, though not necessarily in that order.
Their favorite colors are brown, green, red, black, and yellow, again not necessarily in that order.

The following information is known-

1. Jack's favorite color is red. He is not from India or China.
2. Albert plays football. His favorite color is not yellow, and he is not from the UK.
3. Peter and Sebastian have yellow and green as their favorite colors, not necessarily in that order.
4. Matthew plays badminton. He is from Russia.
5. Peter and Sebastian are from China and UK, not necessarily in that order.
6. Brown is the favorite color of the person from India.
7. Green is the favorite color of the person from China.
8. The person from South Korea plays squash and the person from the UK plays volleyball. One of them has yellow as his favorite

Whose favorite color is black?

Pick **ONE** option

☐ Matthew

26. Standing Order 2

There are five people - Peter, Tyson, Richard, Steve and Quinn - standing in a row. Given the following statements, who is in the middle?

1. Peter is next to Quinn and Steve is next to Richard.
2. Steve is not next to Tyson.
3. Tyson is on leftmost position.
4. Richard is on the second position from the right.
5. Peter stands somewhere to the right of both Quinn and Tyson.
6. Peter and Richard are next to each other.

Pick **ONE** option

☐ Peter

☐ Richard

☐ Steve

☐ Quin

24. Painted Cube

A cube is painted green on all surfaces and is cut into 1000 identical cubes. How many of the smaller cubes have three faces painted?

Pick **ONE** option

☒ 8

☐ 10

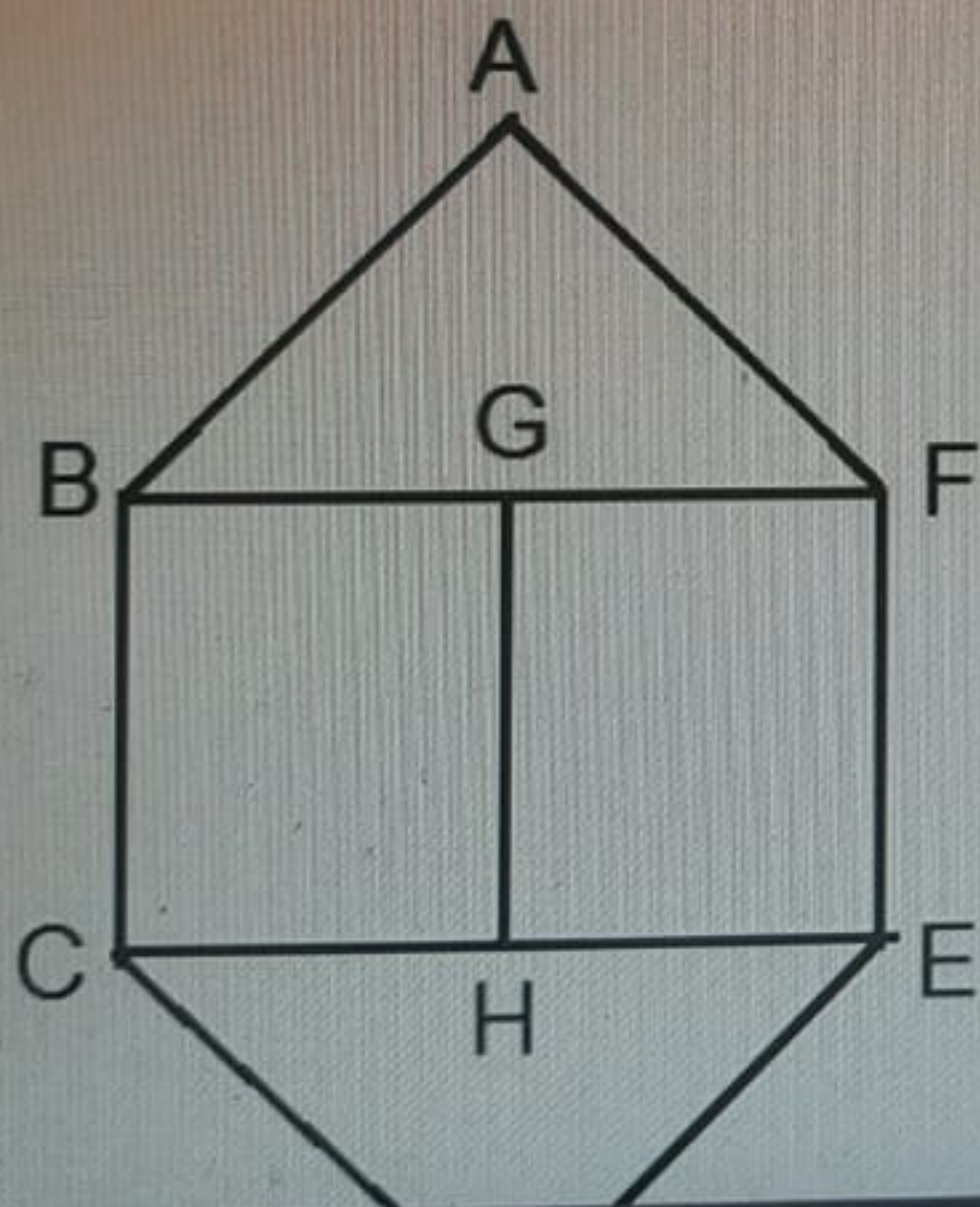
☐ 100

☐ 64

Clear Selection

23. Paths in a Graph 2

There is a network of eight cities (represented by vertices) connected by roads. A traveler must drive on each road segment once, starting and ending points do not have to match. New roads are constructed between C and G, and between H and F. If the journey



Out of the nine ingredients, a chef wants to use exactly five. Given the following statements, how many of the 5 ingredients

The ingredients are apricots, bacon, cake, donuts, eggs, figs, grapes, hazelnuts, and ice cream.

1. If bacon is not used, then grapes, hazelnuts, and ice cream are not used.
2. If apricots are used then bacon and cake are used.
3. If donuts are not used, apricots are not used.
4. Apricots and eggs are used.

Pick **ONE** option

☐ 2

☐ 3

☐ 4

☐ 5